

Computational perspectives on the Classical Decision Problem

Lecture 3: The Two-Variable Fragment

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- The **two-variable fragment**, denoted $\mathcal{FO}^2$, is the fragment of first-order logic over purely relational signatures in which at most two logical variables occur.
- For example:

Every artist who admires a beekeeper despises some dentist

$$\forall x(\text{artst}(x) \wedge \exists y(\text{bkpr}(y) \wedge \text{adm}(y, x)) \rightarrow \exists y(\text{dntst}(y) \wedge \text{desp}(x, y)))$$

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- Observe the re-quantification of the variables x and y in these examples.
- In $\mathcal{FO}^2$, we allow the equality predicate, but we forbid function symbols.

- We can state in $\mathcal{FO}^2$ that a binary relation is **reflexive**, or that it is **symmetric**:

$$\begin{aligned} &\forall x \, r(x, x) \\ &\forall x \forall y (r(x, y) \rightarrow r(y, x)). \end{aligned}$$

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$$\begin{aligned} &\forall x \, r(x, x) \\ &\forall x \forall y (r(x, y) \rightarrow r(y, x)). \end{aligned}$$

- But we cannot obviously state that it is **transitive**:

$$\forall x \forall y \forall z (r(x, y) \wedge r(y, z) \rightarrow r(x, z)).$$

We will see presently that transitivity is not expressible in $\mathcal{FO}^2$.

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- Henceforth, we assume **all predicates are of arity 1 or 2**, and there are **no individual constants** (and certainly no function-symbols).

- A formula of $\mathcal{FO}^2$ is in (Scott) normal form if it has the shape

$$\forall x \forall y (\alpha \vee x = y) \wedge \bigwedge_{h=1}^m \forall x \exists y (\beta_h \wedge x \neq y),$$

where

- m is a positive integer
- $\alpha, \beta_1, \dots, \beta_m$ are quantifier-free, equality-free $\mathcal{FO}^2$ -formulas.

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- $\alpha, \beta_1, \dots, \beta_m$ are quantifier-free, equality-free $\mathcal{FO}^2$ -formulas.
- The first conjunct expresses the universal constraints, and the second, the witness requirements.

Lemma

Let φ be a sentence of $\mathcal{FO}^2$. We can construct, in time bounded by a polynomial function of $\|\varphi\|$, a normal-form $\mathcal{FO}^2$ -formula ψ such that φ and ψ are satisfiable over the same domains of size at least 2.

Stage 1: Let the $\mathcal{FO}^2$ -sentence φ be given. Define $\varphi_0 := \forall x \forall y \varphi$.

Assume φ_i has been defined ($i \geq 0$). Pick an inner-most quantified subformula θ of φ_i , say $\theta = \exists v \chi$, where $\{u, v\} = \{x, y\}$.

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$$\begin{aligned}\psi_{i+1} &:= \forall u (p(u) \leftrightarrow \theta) \\ &\equiv \forall u (p(u) \leftrightarrow \exists v \chi) \\ &\equiv \forall u \exists v (p(u) \rightarrow \chi) \wedge \forall u \forall v (\chi \rightarrow p(u)).\end{aligned}$$

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If $\theta(u) = \forall v \chi$, we proceed similarly.

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$$\psi' := \varphi_n \wedge \psi_n \wedge \psi_{n-1} \wedge \cdots \wedge \psi_1$$

satisfiable over the same domains as φ ; φ_n has the form $\forall x p(x)$, and the ψ_i have the form $\forall u \exists v (p(u) \rightarrow \chi) \wedge \forall u \forall v (\chi \rightarrow p(u))$.

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By trivial logical re-arrangement, ψ' has the form

$$\forall x \forall y \alpha' \wedge \bigwedge_{h=1}^m \forall x \exists y \beta'_h,$$

where α' and the β'_h are quantifier-free.

Stage 2: Observe that $\forall x \forall y \alpha'(x, y)$ is equivalent to

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Thus we obtain the desired form

$$\forall x \forall y (\alpha \vee x = y) \wedge \bigwedge_{h=1}^m \forall x \exists y (\beta_h \wedge x \neq y). \quad \square$$

- Let Σ be a finite, purely relational signature in which all predicates have arity at least 1.
 - A **literal** (over Σ) is an atomic formula or the negation of an atomic formula, e.g.

$$p(x_1), \quad \neg p(x_1), \quad \neg q(x_1, x_2), \quad r(x_1, x_2, x_3, x_4, x_4).$$

- A **k-type** (over Σ) is a maximal consistent set of equality-free literals over Σ involving no variables other than $x_1, \dots, x_k$.

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- A **k -type** (over Σ) is a maximal consistent set of equality-free literals over Σ involving no variables other than $x_1, \dots, x_k$.
- Think of a k -type τ as a complete specification of the relations determined by some structure on a particular selection of k (typically distinct) individuals.

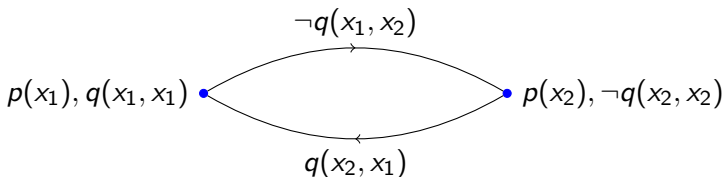
- For example, let the signature Σ be $\{p, q\}$, where p is a predicate of arity 1 and q a predicate of arity 2.
- The following is a 1-type:

$$\pi := \{p(x_1), q(x_1, x_1)\}.$$

- The following is a 2-type:

$$\tau := \{p(x_1), p(x_2), q(x_1, x_1), \neg q(x_1, x_2), q(x_2, x_1), \neg q(x_2, x_2)\}.$$

We might picture τ as follows:



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- In effect, $\text{tp}^{\mathfrak{A}}[\bar{a}]$ is a picture of the substructure obtained by restricting $\mathfrak{A}$ to the subset $\{a_1, \dots, a_k\}$.

- Since Σ is finite, so are k -types over Σ , and we silently identify them with their conjunctions.
- For example, we write the 1-type π and the 2-type τ above as

$$\pi := p(x_1) \wedge q(x_1, x_1)$$

$$\tau := p(x_1) \wedge p(x_2) \wedge q(x_1, x_1) \wedge \neg q(x_1, x_2) \wedge q(x_2, x_1) \wedge \neg q(x_2, x_2).$$

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- For 1-types we usually take the variable to be x (rather than x_1), thus

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- For 2-types we usually take the variables to be x and y (rather than x_1 and x_2), thus:

$$\tau := p(x) \wedge p(y) \wedge q(x, x) \wedge \neg q(x, y) \wedge q(y, x) \wedge \neg q(y, y).$$

- Recall that $\mathcal{FO}^2$ can express the condition that some formula $\psi(x)$ is uniquely instantiated:

$$\exists x \psi(x) \wedge \forall x \forall y (\psi(x) \wedge \psi(y) \rightarrow x = y).$$

- An element of a structure $\mathfrak{A}$ whose 1-type is uniquely instantiated in is called a **king**, and the 1-type in question is called **royal**.
- If $\mathfrak{A}$ interprets a purely relational signature with s predicates of arity at least 1, then the set K kings in $\mathfrak{A}$ satisfies there are at most $|K| \leq 2^s$.

- Suppose $\mathfrak{A} \models \varphi$, where

$$\varphi := \forall x \forall y (\alpha \vee x = y) \wedge \bigwedge_{h=1}^m \forall x \exists y (\beta_h \wedge x \neq y),$$

is a formula in Scott normal-form.

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- Then, for every element $a \in A$, we can choose elements $b_1, \dots, b_m$ from $A \setminus \{a\}$ such that, for all h ($1 \leq h \leq m$)

$$\mathfrak{A} \models \beta_h[a, b_h].$$

We call b_h the h th witness with respect to a .

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- Let K be the set of kings of $\mathfrak{A}$, and suppose m witnesses have been chosen for each of these kings (not necessarily distinct). Then the **court** is the subset of A consisting of the kings and all (chosen) witnesses of kings.

Lemma

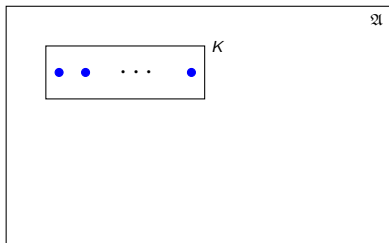
Let φ be a satisfiable $\mathcal{FO}^2$ -formula in Scott normal form

$$\forall x \forall y (\alpha \vee x = y) \wedge \bigwedge_{h=1}^m \forall x \exists y (\beta_h \wedge x \neq y),$$

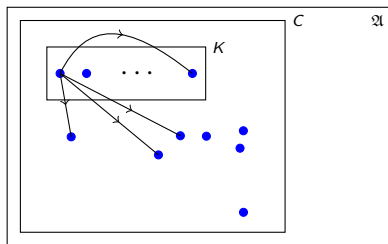
over a signature Σ of unary and binary predicates. Then φ has a model of size at most $3m \cdot 2^{|\Sigma|}$.

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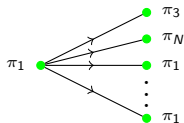
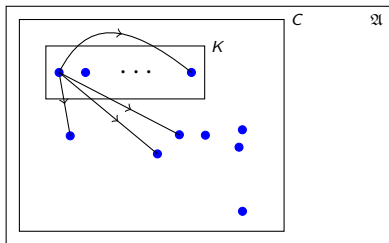
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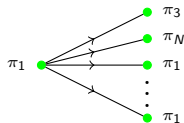
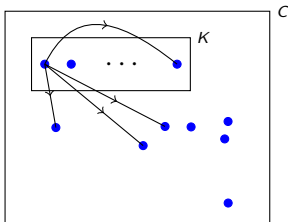
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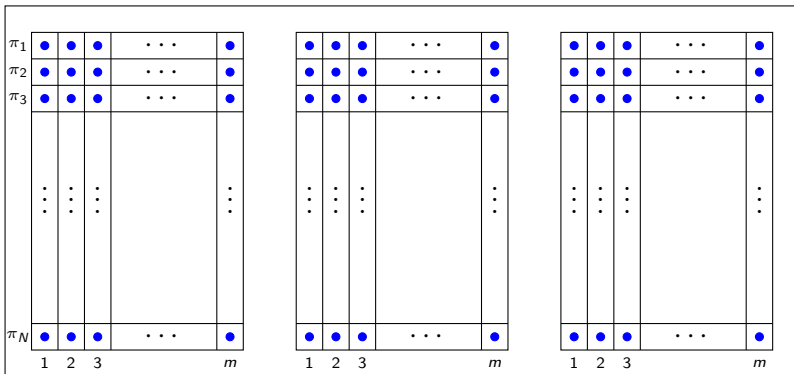
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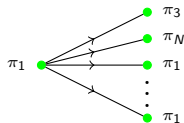
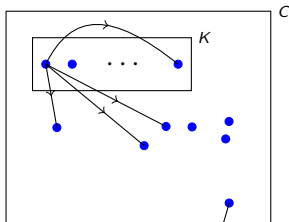


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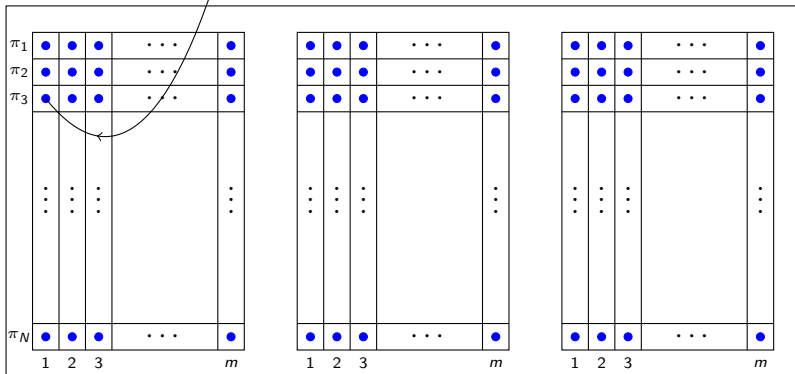


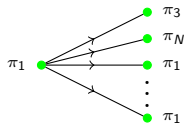
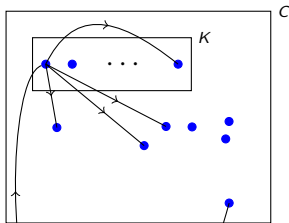
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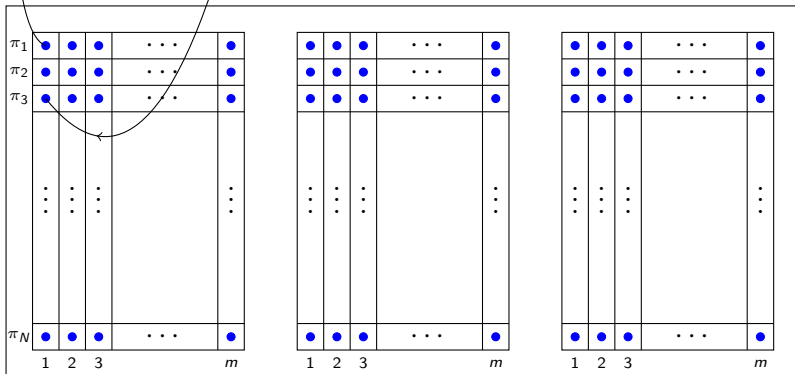


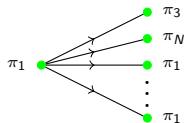
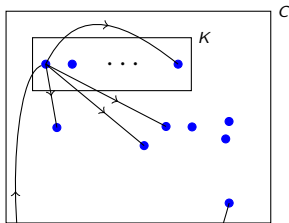
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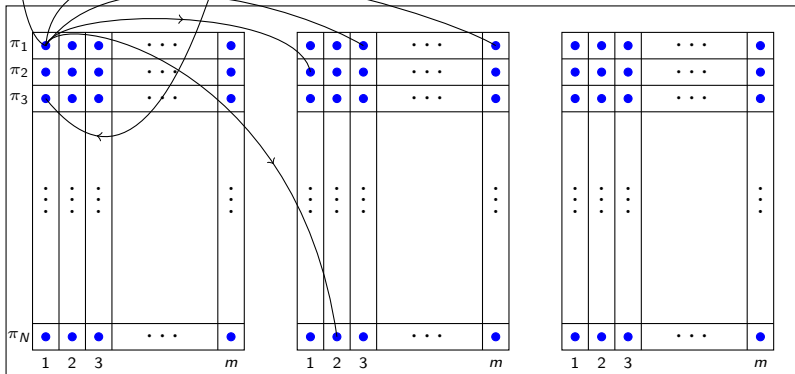


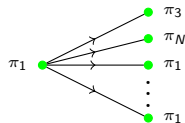
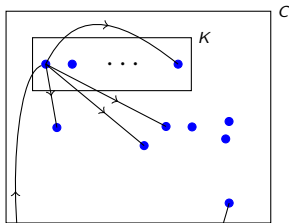
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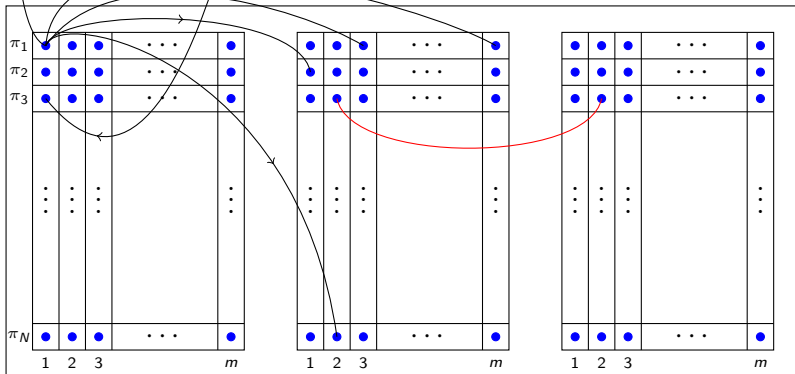


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Corollary

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Corollary

The fragment $\mathcal{FO}^2$ has the finite model property.

- It is possible to obtain a matching lower bound for the complexity of $\text{Sat}(\mathcal{FO}^2)$, specifically:

Lemma

The problem $\text{Sat}(\mathcal{FO}^2)$ is NEXPTIME-hard.

Proof.

The $\mathcal{FO}_{\text{Mon}}$ -formula constructed in the previous lecture happens to be in $\mathcal{FO}^2$. □

- Thus we have

Theorem

The problem $\text{Sat}(\mathcal{FO}^2)$ is NEXPTIME-complete.

- We mentioned earlier that it did not seem to be possible to state in $\mathcal{FO}^2$ that a binary relation t is **transitive**:

$$\forall x \forall y \forall z (r(x, y) \wedge r(y, z) \rightarrow r(x, z)).$$

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- We mentioned earlier that it did not seem to be possible to state in $\mathcal{FO}^2$ that a binary relation t is **transitive**:

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But perhaps there is some clever way of expressing this in $\mathcal{FO}^2$ after all ...

- The finite model property shows that there is not. If the predicate t is interpreted as a transitive relation, then

$$\forall x \exists y t(x, y) \wedge \forall x \neg t(x, x)$$

is satisfiable, but only in infinite structures:



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- We observed that a matching lower bound can be proved. Hence $\text{Sat}(\mathcal{FO}^2)$ is NEXPTIME-complete.
- We used the FMP to show that transitivity is not expressible in $\mathcal{FO}^2$.